

Arduino Timer Interrupts by

amandaghassaei (/member/amandaghassaei/) in arduino (/tag/type-id/category-technology/channel-arduino/)

Download

 (/id/Arduino-Timer-Interrupts/?ALLSTEPS)

6 Steps

▶ (/id/Arduino-Timer-Interrupts/step5/Example-3-DAC/)

+ Collection

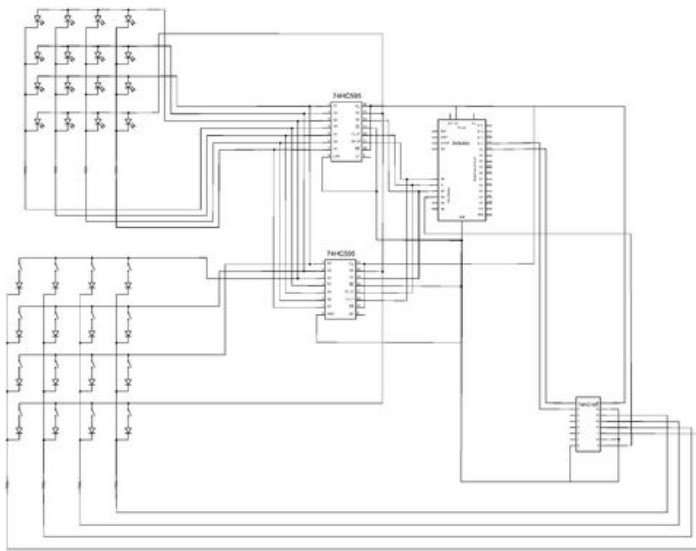
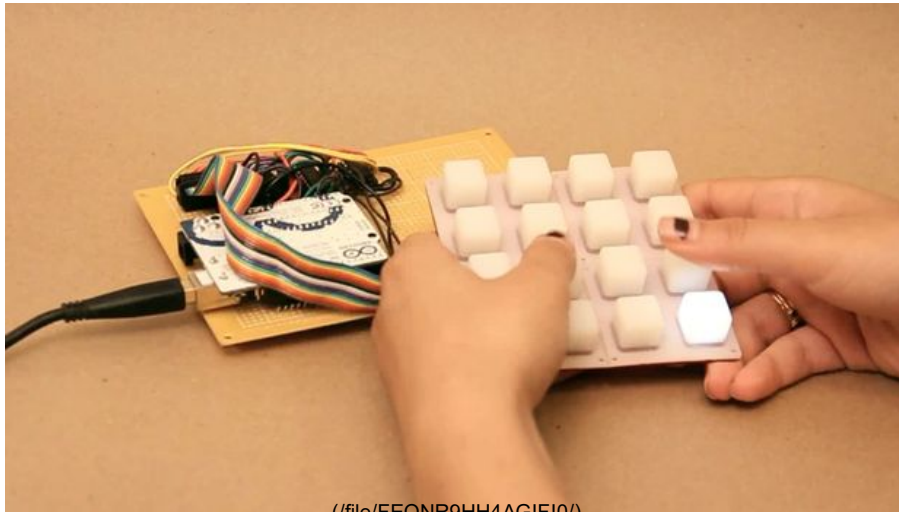
I Made it!

♥ Favorite

🔗 Share ▼



Step 4: Example 2: Serial Communication

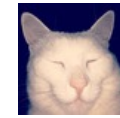


About This Instructable

 779.990 views

♥ 537 favorites

License:

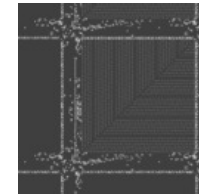


amandaghassaei
(/member/amandaghassaei/)
uh-man-duh-guss-eye-dot-com
(<http://www.amandaghassaei.com>)

(/member/amandaghassaei/) 3362

Bio: I'm a grad student at the Center for Bits and Atoms at MIT Media Lab. Before that I worked at Instructables, writing code for ... More » (/member/amandaghassaei/)

More by amandaghassaei:



(/id/OTCA-

Metapixel-Conways-Game-of-Life/)



(/id/9-Degrees-of-

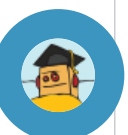
Freedom-IMU/)



Freedom-IMU/)

(/id/Twitter-Controlled-Pet-Feeder/)

This project is a 4x4 backlit button pad. The project connects to my computer via usb, it sends information about the buttons to the computer and receives information about how to light up the LEDs. Here is a video:





For this project, I used timer2 interrupts to periodically check if there was any incoming serial data, read it, and store it in the matrix "ledData[]". If you take a look at the code you will see that the main loop of the sketch is what is actually responsible for using the info in ledData to light up the correct LEDs and checking on the status of the buttons (a function called "shift()"). The interrupt routine is as short as possible- just checking for incoming bytes and storing them appropriately.

Here is the setup for timer2:

```
cli();//stop interrupts
//set timer2 interrupt every 128us
TCCR2A = 0;// set entire TCCR2A register to 0
TCCR2B = 0;// same for TCCR2B
TCNT2 = 0;//initialize counter value to 0
// set compare match register for 7.8khz increments
OCR2A = 255;// = (16*10^6) / (7812.5*8) - 1 (must be <256)
// turn on CTC mode
TCCR2A |= (1 << WGM21);
// Set CS21 bit for 8 prescaler
TCCR2B |= (1 << CS21);
// enable timer compare interrupt
TIMSK2 |= (1 << OCIE2A);
sei();//allow interrupts
```

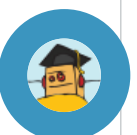
Here's the complete Arduino sketch:

```
//BUTTON TEST w/ 74HC595 and 74HC165 and serial communication
//by Amanda Ghassaei
//June 2012
//https://www.instructables.com/id/Arduino-Timer-Interrupts/

/*
 * This program is free software; you can redistribute it and/or modify
 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation; either version 2 of the License, or
 * (at your option) any later version.
 *
 */

//this firmware will send data back and forth with the maxmsp patch "beat slicer"
```

download the MaxMSP (<http://cycling74.com/downloads/>) patch below (it will run in Max Runtime as well).  Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)





beat slicer.zip

Download (<https://cdn.instructables.com/ORIG/FFP/TBHG/H4AGIFKT/FFPTBHG4AGIFKT.zip>)
(<https://cdn.instructables.com/ORIG/FFP/TBHG/H4AGIFKT/FFPTBHG4AGIFKT.zip>)

« Previous (/id/Arduino-Timer-Interrupts/step3/Example-1-Bike-Speedometer/)

Next » (/id/Arduino-Timer-Interrupts/step5/Example-3-DAC/)

Download

View All Steps (/id/Arduino-Timer-Interrupts/?ALLSTEPS)



We have a be nice comment policy.
Please be positive and constructive.

I Made it!

Add Images

Post Comment



AravindP4 (/member/AravindP4)

8 days ago

Reply

thank u sir



andreondra (/member/andreondra)

5 months ago

Reply

Great instructable! Thank you very much!



MalachiB1 (/member/MalachiB1)

a year ago

Reply

Thank you for breaking this down. I've read a lot of articles about timers but only yours makes it clear!



dineshc31 (/member/dineshc31) ▶ **MalachiB1** (/member/MalachiB1)

Reply

how exactly this is working for you all ?

6 months ago

For me while compiling in arduino 1.6.10 it is showing

error: 'TIMSK1' was not declared in this scope which is same for all above registers. do we need any #include header file here?



KennyL15 (/member/KennyL15)

9 months ago

Reply

Should also mention that the clock prescaler bit names CSX0, CSX1, CSX2 has the first number as 'X'. That number 'X' is linked to the timer number. So if 'X' is 0, then CS00, CS01, CS02 are prescaler bit names for timer 0; and the second value represents the bit position. So C00 means bit 0 (right-most-bit) of the 8-bit register TCCR0B, which is for timer 0.

The reason why I'm indicating this is because the tables (diagram) that go with "prescalers and compare match register" is confusing - since there's no mention about what CS10, CS11, CS12 etc are actually related to.



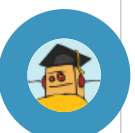
KennyL15 (/member/KennyL15)

9 months ago

Reply

"so a counter will take 10/16,000,000 seconds to reach a value of 9 (counters are 0 indexed)"

✕ Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)



Do you actually mean it takes 10/16000000 seconds to begin (initial state) from index 0 and get back to a 0 index again? That is, initial index of 0. Then first increment is 1, second increment is 2, third increment is 3, all the way to 9th increment to index 9, and then one extra increment to get back to index 0 again. Altogether 10 increments to 'get back to 0'.



PeterMortensen (/member/PeterMortensen)

11 months ago

[Reply](#)

The comment for the line containing "ISR(TIMER2_COMPA_vect)" is confusing. "**timer1**" should probably be "**timer2**".



Sharmila07 (/member/Sharmila07)

a year ago

[Reply](#)

my signal is in the frequency range between 20Hz and 500Hz, so i am taking the sampling frequency as 5000Hz (sampling time will be .2ms), after sampling i need to store the sampled values in the SD card and so i need to use spi. i want to know how much time will spi take to store in SD card and will i be able to sample my signal at this rate? kindly help



ParthJ5 (/member/ParthJ5)

a year ago

[Reply](#)

Can anyone please explain the connections to arduino in example 1 ?
I want to know how is the signal generator and Oscilloscope is connected to the Arduino ?

Thanks



Bruce McCarthy (/member/Bruce McCarthy)

a year ago

[Reply](#)

Brilliant! I struggled with the datasheets for a few days before finding this instructable, and now it's all clear.

Thanks for sharing.



avionics2 (/member/avionics2)

a year ago

[Reply](#)

This is the best interrupts explanations I have come across.

Thanks !



jerrychee94 (/member/jerrychee94)

a year ago

[Reply](#)

Hi, currently im doing project where it prompt user in serial communication and doing interrupt ISR at the same time. So , when user enter a specific value , it will stop the interrupt for few seconds and then restart the interrupt again. However , when restart the interrupt , the timing is out.(expected interrupt time : 5 microsec per interrupt , actual interrupt time : higher than 5 microsec)

my code is here..

```
void setup()
```

```
{
```

```
cli();
```

```
TCCR0A = 0;
```

```
TCCR0B = 0;
```

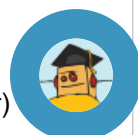
```
TCNT0 = 0;
```

```
OCR0A = 80;// = (16*10^6) / (200000*1) - 1 (must be <256)
```

```
//200000hz=5microsec
```

```
TCCR0A |= (1 << WGM01);
```

```
TCCR0B |= (0 << CS02)|(0 << CS01)|(1 << CS00);
```



```

TIMSK0 |= (1 << OCIE0A);

sei();

}

void loop()
{
  if(serial.available>0)
  {
    //if user press something...

    TIMSK0 &= ~(1 << OCIE0A);
    delay(10000);
    TIMSK0 |= (1 << OCIE0A);
  }
}

ISR(TIMER0_COMPA_vect)
{
  //code
}

```



PiraC (/member/PiraC)

a year ago

Reply

Good Work. Thank you!



sam.gwilliam (/member/sam.gwilliam)

a year ago

Reply

```

// turn on CTC mode
TCCR0A |= (1 << WGM01);
// Set CS01 and CS00 bits for 64 prescaler
TCCR0B |= (1 << CS01) | (1 << CS00);

```

What is the TCCR0A register? TCCR0B must be the 3-bit one as illustrated (where you set the prescaler) but I don't recall any mention of the other one.



MichealM1 (/member/MichealM1)

a year ago

Reply

This made it much clearer than other reads I have been to. Based on this explanation, I was able to devise a routine to discipline my software clock based on the 16MHz crystal. One of my units (Uno) is ticking at a very wrong speed. Gaining about 30 seconds per day. That's a LOT! But the software can now take care of it and any other errant crystal it encounters. Thanks!!



MichealM1 (/member/MichealM1) ▶ MichealM1 (/member/MichealM1)

Reply

a year ago

Sorry. Left out that I am getting NTP time hacks once per day but losing 30 seconds per day is way overboard. Maybe I will replace the crystal and see but as long as I can correct for it, it's not so bad.



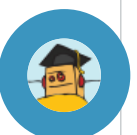
Baguda (/member/Baguda) made it!

a year ago

Reply

In the code you set the timer1 register to CTC mode by "TCCR1B |= (1 << WGM12);". However, for all of the examples using timer0 and timer2, you set the CTC mode by "TCCR0B |= (1 << WGM01);" why do you set bit1 high instead of bit2 for timers 0 and 2 to put them into CTC mode? I found this table online that lists the mode settings(see attached), and it states that (100) is CTC mode. However, it is only for timer1. Is it different for timers 0 and 2?

Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)



Generation III Design(s)						
Reactor	Reactor Type	Reactor Power	Estimated Core Burnup	WGR	WGR Type	WGR Power
0	0	0	Normal		QWTR	0
0	0	1	PRM, Phase Control, 850		QWTR	75
0	1	0	PRM, Phase Control, 850		QWTR	75
0	1	1	PRM, Phase Control, 1030		QWTR	80
0	0	0	CTC		QWTR	0
0	0	1	Fast PRM, 850		QWTR	80
0	1	0	Fast PRM, 850		QWTR	80
0	1	1	Fast PRM, 1030		QWTR	80
0	0	0	PRM, Phase and Frequency Control		KRM	0
0	0	1	PRM, Phase and Frequency Control		QWTR	80
0	1	0	PRM, Phase Control		KRM	75
0	1	1	PRM, Phase Control		QWTR	80
1	0	0	CTC		KRM	0
1	0	1	Slow PRM		KRM	75
1	0	0	Fast PRM		KRM	80
1	0	1	Fast PRM		QWTR	80

(<https://cdn.instructables.com/FK7/FFSP/IJIUVOFI/FK7FFSPIJIUVOFI.LARGE.jpg>)



AnthonyP7 (/member/AnthonyP7)

2 years ago

Reply

If the Arduino is executing some code within an interrupt does the counter continue to count up so that the next interrupt is correctly timed?

sandro2204 (/member/sandro2204) ▶ **AnthonyP7 (/member/AnthonyP7)**

Reply

a year ago

Yes it does. But you should keep your interrupt service routine (ISR) as short as possible. Store a variable, set a flag and that's it. Do time consuming stuff in your main loop.

hal4512 (/member/hal4512)

a year ago

Reply

according to datasheet of atmega328p p106 -108 when we choose values of COM0A0 and COM0B0 for timer0

we set the behavior of pins OC0A OC0B After having config them as output and hardware will proceed as its describe in the datasheet.

all modes Normal mode ,CTC mode non PWM ..ect ..are described in page 100

when you'll read:For generating a waveform output in CTC mode the OC0A output can be set to toggle each logical level on each compare match (you don't have to toggle it by soft).

hal4512 (/member/hal4512)

a year ago

Reply

you don't need to write a code for toggling because atmega328p has 3 timers

and every timer has 2 output pins ex: timer2 has OC2A pin 11 and OC2B pin3
and in isr routine u can do other thing.



KevinK91 (/member/KevinK91) ▶ **hal4512 (/member/hal4512)** a year ago

Reply

could you please elaborate more on why the toggle code isn't necessary in this sketch?



hackerh (/member/hackerh)

a year ago

Reply

so if you wanted an interrupt every second (frequency of 1Hz):

compare match register = $[16,000,000 / (\text{prescaler} * 1)] - 1$
with a prescaler of 1024 you get:



metrog (/member/metrog)

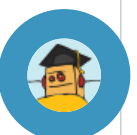
a year ago

Reply

Hi,

Your project and explanations helped me a lot starting with my project. Thanks!

I'm using Arduino uno timer 0 and 2 to output 2 distinct (in sync) MIDI Clock signal and it works like charm. I can change BPM with a knob going from 30 to 250.



I'm struggling on my next step which is introducing an offset (-200ms / +200ms) between the 2 signals using an analogRead on a knob.

Would you know how to induce such an offset for a Serial output?

Timr0 and Timer2 doesnt have much room to change start point and end point and the induced delay would only be a few milliseconds maybe.

Well if this message finds you and you wish to reply, I'm all ears opened.

Thanks again,

Stef



tejonbiker (/member/tejonbiker)

a year ago

Reply

Wonderful, I worked with PICs but I see the AVR work slightly different



youraagh (/member/youraagh)

2 years ago

Reply

Hi, great tutorial, thanks heaps.

What libraries do I need to include, I'm getting compiler errors saying it doesn't know the names of the timing registers.



PravinahS (/member/PravinahS)

2 years ago

Reply

Hi Amanda,

I found these article very informative. I'm designing a card reader and i have included these timer interrupts in my programs.I have several confusions to use these timer interrupts on Arduino Leonardo.I have attached pins on pin 3 and pin 2 for D0 and D1 from my Mifare card reader.My card is a Mifare (32bits) type, so when i batch these card,i'm getting the extra bit of 33,so i decided to use timer interrupts just to execute exact 32 bits,but still i'm getting these extra bit.Please help me..Im lost!!



SitthinutK (/member/SitthinutK)

2 years ago

Reply

Thanks!!



RosmiR (/member/RosmiR)

2 years ago

Reply

In an Arduino code for pulse sensor 256 prescalar is activated by TCCR2b=0x06 . Why is it so ? Why is it not 0x04 as per the table for Timer0. Is it possible to make an interrupt of 100 Hz using Timer 2. (ocr2a <256, 1024 prescalar is not present). Could you help me with this. I am just a beginner.



RosmiR (/member/RosmiR)

2 years ago

Reply

How could I make an interrupt of 100Hz with timer 2. Could you share the table for timer2 similar to the one for Timer0 .



ProfRobNewton (/member/ProfRobNewton)

2 years ago

Reply

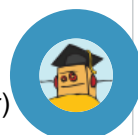
Just wondering if it would not be easier and more accurate to generate an interupt when a switch change is registered and use millis() to determine the time since the last switch change? Sampling every 1ms using the method above may miss a switch change??



andycomiC (/member/andycomiC) ▶ **ProfRobNewton (/member/ProfRobNewton)**



Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)



I am no expert but to my knowledge the problem with switch state interrupts is that they bounce and you can't denounce a switch using software on an interrupt routine as its already been triggered (as in the interrupt routine is already done before you have a chance to de-bounce.). The other way is to use a hardware based button debounce switch which just uses a capacitor. This will solve the problem. However unless your project is huge, polling a switch every x cycles doesn't really make any difference to any project.



Fardenco (/member/Fardenco) ▶ andycomic (/member/andycomic)

Reply

2 years ago

I had the problem with a motor encoder, just put a 10k Ohms resistor to Vcc and a 1µF capacitor to GND around your interrupt pin, this will be ok !!

Pin interrupt would be more efficient because you'll never miss a change, and above that, you'll not waste time with useless interruption =)



DavidS13 (/member/DavidS13) ▶ ProfRobNewton (/member/ProfRobNewton)

Reply

2 years ago

While I don't think sampling every 1ms may miss a switch change (think about it, 1000 times a second???) I too would like to know if the more efficient way would be to make an interrupt for the switch change. I have a project that would use something similar and I'm trying to determine the most efficient course of action.



manoch_dc (/member/manoch_dc)

2 years ago

Reply

thank for Example interrupts that working.

!!But void loop() has not work

my increase program

```
void setup(){
```

```
//set pins as outputs
```

```
pinMode(7, OUTPUT); //my led
```

```
pinMode(8, OUTPUT);
```

```
pinMode(9, OUTPUT);
```

```
pinMode(13, OUTPUT);
```

```
cli();//stop interrupts
```

same example

```
sei();//allow interrupts
```

```
}
```

```
void loop(){
```

```
digitalWrite(7, HIGH); // turn the LED on (HIGH is the voltage level)
```

```
delay(100); // wait for a second
```

```
digitalWrite(7, LOW); // turn the LED off by making the voltage LOW
```

```
delay(100); // wait for a second
```

```
//do other things here
```

```
}
```



coltenschlosser (/member/coltenschlosser)

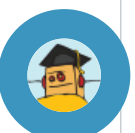
2 years ago

Reply

I made a python script that walks you through setting up the code for timer interrupts.

<https://gist.github.com/cltnschlosser/e747efae07a76e117e02>

✕ Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)





killercatfish (/member/killercatfish)

2 years ago

[Reply](#)

Could this be used for a 2 player reflex timer game? Trying to push your players button on multiples of 5 seconds and getting a point when you do.



zinedine (/member/zinedine)

2 years ago

[Reply](#)

Good job, I want to import 200khz frequency, it's about 5us, but I only have 13.6us with every OCRnA that I used to change from 0 to 80. I'm using 2560. How can I do that?

Thanks



faizal91 (/member/faizal91)

2 years ago

[Reply](#)

Thank you... Very nice tutorial!



EnekoL (/member/EnekoL)

2 years ago

[Reply](#)

Hei, very usefull application, but i've one cuestion, how can i call de interrump for timer 1 inside the loop??Thanks!



ohoilett (/member/ohoilett) ▶ EnekoL (/member/EnekoL)

2 years ago

[Reply](#)

I could be wrong. You may want to check

<http://arduino.cc/en/Reference/AttachInterrupt>



ohoilett (/member/ohoilett) ▶ EnekoL (/member/EnekoL)

2 years ago

[Reply](#)

You don't call interrupts like normal functions. An interrupt service routine (ISR) is "called" or better yet "triggered" at the specified frequency that you determined when you set up the interrupt.



thegoodhen (/member/thegoodhen)

3 years ago

[Reply](#)

Hello... The code is very useful, however, be aware that the sketch you posted doesn't seem to ever get into the main loop!



DavidS13 (/member/DavidS13) ▶ thegoodhen (/member/thegoodhen)

[Reply](#)

2 years ago

I'm kind of a noob about all this, but unless I'm mistaken the point of interrupts is that you don't NEED a main loop. The main loop just runs forever and everything is done when interrupts happen. It is an extremely efficient way to code....



Unidentified Identity (/member/Unidentified Identity)

3 years ago

[Reply](#)

can we generate square wave with varying pulse width with this interrupt ?



amandaghassaei (/member/amandaghassaei) (author) ▶ Unidentified Identity

(/member/Unidentified Identity)

3 years ago

[Reply](#)

yes it's possible, but probably easier to just use the analogWrite command

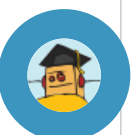


srinivas_arduino (/member/srinivas_arduino)

3 years ago

[Reply](#)

✕ Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)



sir please send program for generating 15000 sample values in 10 sec using timer interrupt values adc values from accelerometer sensor



srinivas_arduino (/member/srinivas_arduino)

3 years ago

Reply

sir please send program for generating 15000 sample values using timer interrupt values adc values from accelerometer sensor



AbdulW87 (/member/AbdulW87)

3 years ago

Reply

i am confused about something in the code, the line

// turn on CTC mode

TCCR0A |= (1 << WGM01);

shouldnt it be TCCR0A |= (1<<WGM02); or am i understanding this wrong, i am a beginner at arduino so its still very confusing to me.



O-Zo (/member/O-Zo)

3 years ago

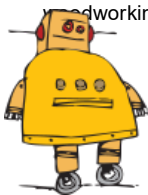
Reply

Thank you for this tutorial, I'm struggling with a project and this helped me stumble at least to the right direction.

↓ More Comments

FEATURED CHANNELS

Woodworking (/tag/type-id/category-workshop/channel-woodworking/)	Paper (/tag/type-id/category-craft/channel-paper/)	Kitchen Hacks (/tag/type-id/keyword-kitchen%20hacks/?sort=FAVORITES)	Puzzles (/tag/type-id/category-play/channel-puzzles/)	Laser Cutting (/tag/type-id/category-workshop/channel-laser-cutting/)	Space (/tag/type-id/keyword-astronomy/?sort=FAVORITES)	Homesteading (/tag/type-id/category-home/channel-homesteading/)	3D Printing (/tag/type-id/category-technology/channel-3D-Printing/)	Sewing (/tag/type-id/category-craft/channel-sewing/)
--	---	---	--	--	---	--	--	---



Newsletter

Join 2 million + to receive instant DIY inspiration in your inbox.

enter email

I'm in!

About Us

Who We Are (/about/)

Advertise (/advertise/)

Contact (/about/contact.jsp)

Jobs (/community/Positions-available-at-Instructables/)

Help (/id/how-to-write-a-great-instructable/)

Resources

For Teachers (/teachers/)

Artists in Residence (/air)

Gift Premium Account (/account/give?source=footer)

Forums (/community/)

Answers (/tag/type-question/?sort=RECENT)

Sitemap (/sitemap/)

Find Us

Facebook (<http://www.facebook.com/instructables>)

Youtube (<http://www.youtube.com/user/instructablestv>)

Twitter (<http://www.twitter.com/instructables>)

Pinterest (<http://www.pinterest.com/instructables>)

Google+ (<https://plus.google.com/+instructables>)

Terms of Service (<http://usa.autodesk.com/adsk/servlet/item?siteID=123112&id=21959721>) |

Privacy Statement (<http://usa.autodesk.com/adsk/servlet/item?siteID=123112&id=21292079>) |

Legal Notices & Trademarks (<http://usa.autodesk.com/legal-notices-trademarks/>) | Mobile Site (<https://www.instructables.com>)



(<http://usa.autodesk.com/adsk/servlet/pc/index?id=20781545&siteID=123112>)

© 2016 Autodesk, Inc.

✕ Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)

